

Data wrangling fundamentals

What do you do with data?

- Data manipulation and cleaning
- Calculate summary statistics
- Visualization
- Input for modeling

Data manipulation

```
1 glimpse(starwars)
```

Rows: 87

Columns: 14

```
$ name      <chr> "Luke Skywalker", "C-3P0", "R2-D2", "Darth Vader",  
"Leia Or...  
$ height    <int> 172, 167, 96, 202, 150, 178, 165, 97, 183, 182, 188,  
180, 2...  
$ mass      <dbl> 77.0, 75.0, 32.0, 136.0, 49.0, 120.0, 75.0, 32.0,  
84.0, 77...  
$ hair_color <chr> "blond", NA, NA, "none", "brown", "brown, grey",  
"brown", N...  
$ skin_color <chr> "fair", "gold", "white, blue", "white", "light",  
"light", "...  
$ eye_color  <chr> "blue", "yellow", "red", "yellow", "brown", "blue",
```

What manipulation might I want to do with the `starwars` data?

dplyr: Tools for data wrangling



- part of the tidyverse
- provides a “grammar of data manipulation”: useful verbs (functions) for manipulating data
- we will cover the key dplyr functions

Some core verbs for data wrangling

- `filter()`: take a subset of the rows (i.e., observations)
- `select()`: take a subset of the columns (i.e., features, variables)
- `mutate()`: add or modify existing columns
- `arrange()`: sort the rows
- `group_by()`: group rows by one or more variables
- `summarize()`: aggregate the data across rows (often after grouping)

Creating a subset of the rows

Question: Suppose I only want the droids in the `starwars` data. How would I choose only those rows?

Creating a subset of the rows

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```
1 filter(starwars, species == "Droid")
```

```
# A tibble: 6 × 14
```

	name	height	mass	hair_color	skin_color	eye_color	birth_year	sex	gender
	<chr>	<int>	<dbl>	<chr>	<chr>	<chr>	<dbl>	<chr>	<chr>
1	C-3P0	167	75	<NA>	gold	yellow	112	none	masculi...
2	R2-D2	96	32	<NA>	white, blue	red	33	none	masculi...
3	R5-D4	97	32	<NA>	white, red	red	NA	none	masculi...
4	IG-88	200	140	none	metal	red	15	none	masculi...

Creating a subset of the rows

```
1 starwars |>
2   filter(species == "Droid")
```

```
# A tibble: 2 × 14
```

	name	height	mass	hair_color	skin_color	eye_color	birth_year	sex
	<chr>	<int>	<dbl>	<chr>	<chr>	<chr>	<dbl>	<chr>
	<chr>							
1	C-3P0	167	75	<NA>	gold	yellow	112	none
	masculine							
2	R2-D2	96	32	<NA>	white, blue	red	33	none
	masculine							

```
# i 5 more variables: homeworld <chr>, species <chr>, films <list>,  
#   vehicles <list>, starships <list>
```


Calculating summary statistics

Question: What is the average height for droids in the dataset?

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```
1 starwars |>
2   filter(species == "Droid") |>
3   summarize(mean_height = mean(height))
```

```
# A tibble: 1 × 1
  mean_height
    <dbl>
1         NA
```

- pipes (`|>`) can be chained together
- `summarize` calculates summary statistics
- Why am I getting NA?

Handling missing values

```
# A tibble: 6 × 14
```

```
  name    height  mass hair_color skin_color eye_color birth_year sex
  <chr>    <int> <dbl> <chr>      <chr>      <chr>      <dbl> <chr>
1 C-3P0    167    75 <NA>      gold        yellow      112 none
masculi...
2 R2-D2     96    32 <NA>      white, blue red        33 none
masculi...
3 R5-D4     97    32 <NA>      white, red  red        NA none
masculi...
4 IG-88    200   140 none      metal        red        15 none
masculi...
```

```
1 starwars |>
2   filter(species == "Droid") |>
3   summarize(mean_height = mean(height, na.rm=T))
```

```
# A tibble: 1 × 1
```

```
  mean_height
  <dbl>
1      131.
```

Calculating summary statistics

Question: What if I want the average height for *humans*?

```
1 starwars |>
2   filter(species == "Droid") |>
3   summarize(mean_height = mean(height, na.rm=T))
```

Calculating summary statistics

Question: What if I want the average height for *humans*?

```
1 starwars |>
2   filter(species == "Human") |>
3   summarize(mean_height = mean(height, na.rm=T))
```

```
# A tibble: 1 × 1
  mean_height
    <dbl>
1         178
```

Calculating summary statistics

Question: What is the average height for *each* species?

Calculating summary statistics

Question: What is the average height for *each* species?

```
1 starwars |>
2   group_by(species) |>
3   summarize(mean_height = mean(height, na.rm=T))
```

A tibble: 38 × 2

	species <chr>	mean_height <dbl>
1	Aleena	79
2	Besalisk	198
3	Cerean	198
4	Chagrian	196
5	Clawdite	168
6	Droid	131.
7	Dug	112
8	Ewok	88
9	Geonosian	183
10	Gungan	209.

Creating new variables

Question: What is the distribution of the ratio of body mass to height?

Creating new variables

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```
1 starwars |>  
2   mutate(body_ratio = mass/height)
```

Creating new variables

```
1 starwars |>
2   mutate(body_ratio = mass/height) |>
3   group_by(species) |>
4   summarize(mean_ratio = mean(body_ratio, na.rm=T),
5             sd_ratio = sd(body_ratio, na.rm=T))
```

A tibble: 38 × 3

	species <chr>	mean_ratio <dbl>	sd_ratio <dbl>
1	Aleena	0.190	NA
2	Besalisk	0.515	NA
3	Cerean	0.414	NA
4	Chagrian	NaN	NA
5	Clawdite	0.327	NA
6	Droid	0.453	0.174
7	Dug	0.357	NA
8	Ewok	0.227	NA
9	Geonosian	0.437	NA
10	Gungan	0.351	0.0207

Creating new variables

```
1 starwars |>
2   mutate(body_ratio = mass/height) |>
3   group_by(species) |>
4   summarize(mean_ratio = mean(body_ratio, na.rm=T),
5             sd_ratio = sd(body_ratio, na.rm=T),
6             N = n())
```

A tibble: 38 × 4

	species <chr>	mean_ratio <dbl>	sd_ratio <dbl>	N <int>
1	Aleena	0.190	NA	1
2	Besalisk	0.515	NA	1
3	Cerean	0.414	NA	1
4	Chagrian	NaN	NA	1
5	Clawdite	0.327	NA	1
6	Droid	0.453	0.174	6
7	Dug	0.357	NA	1
8	Ewok	0.227	NA	1
9	Geonosian	0.437	NA	1
10	Gungan	0.351	0.0207	3

Summary so far

- `filter`: choose certain rows
- `summarize`: calculate summary statistics
- `group_by`: group rows together
- `mutate`: create new columns

Class activity

- Work with a neighbor on the class activity (linked on Canvas and course website)
- At the end of class, submit your work as an HTML file on Canvas (one per group, list all your names)
- I will come around and answer any questions

For next time, read:

- Chapter 3 in *R for Data Science* (2nd ed.)
- Chapter 4 in *Modern Data Science with R*